

Understand NX-OS OSPF NSSA Type-7 LSA Forwarding Address Selection

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Introduction

This document extends the concepts described in another document [Understand Selection of Forwarding Address in OSPF](#) and explains the difference when you use NX-OS, the network operating system for Cisco Nexus-series Ethernet switches.

Explanation

According to above document, the forwarding address is selected on Autonomous System Border Router (ASBR) using these rules:

1. If there is a loopback interface configured in the area, then IP address of the loopback interface is selected as Forwarding Address (FA).
2. If the first condition is not met, then IP address of the first interface on the Open Shortest Path First (OSPF) interface list is selected as a forwarding address. You can see OSPF interface list by using **show ip ospf interface brief** command. The interface on top will be the last interface which was attached to OSPF.

However, the second rule is not applicable to NX-OS.

In NX-OS, when there is no loopback interface, OSPF interface ID determines the forwarding address.

IP address of the interface with the highest ID is selected as FA, regardless of whether the interface is passive or not.

Please note that interface ID is dynamic and may change after the system is reloaded.

If an interface is removed from OSPF and is added back, its ID in OSPF interface list may change if another new interface is added into OSPF between the delete and add operations.

The reason is that OSPF interface ID resource is used cyclically.

When an interface is removed from OSPF interface list, the assigned interface ID will be recycled, i.e., when another interface is added to OSPF interface, the ID will be assigned to the interface.

Only when all recycled IDs are depleted, a new interface ID will be allocated for the next OSPF interface.

An example is provided:

```
show ip ospf interface brief vrf A OSPF Process ID 10 VRF A Total number of interface: 4
Interface ID Area Cost State Neighbors Status Vlan101 452 0.0.0.7 5 BDR 2 up Vlan201 678 0.0.0.7
100 DR 0 up <----- highest ID = FA Vlan301 160 0.0.0.7 100 DR 0 up Po11 247 0.0.0.7 1000 P2P 1
up
```

- When vlan201 is removed from OSPF, ID 678 is freed.
- Then, when van 202 is added to OSPF, ID 678 is assigned to vlan 202.
- Then vlan 201 is added back, a new larger ID is assigned to vlan 201.

```
show ip ospf interface brief vrf B OSPF Process ID 10 VRF B Total number of interface: 4
Interface ID Area Cost State Neighbors Status Vlan102 490 0.0.0.7 5 BDR 2 up <----- highest ID
= FA Vlan202 51 0.0.0.7 100 DR 0 up Vlan302 102 0.0.0.7 100 DR 0 up Po12 275 0.0.0.7 1000 P2P 1
up interface Vlan101 vrf member A ip address 192.168.1.1/24 <-----FA interface Vlan102 vrf
member B ip address 10.1.1.1/24 <-----FA N7K#show ospf data nssa-external172.16.1.140 det vrf
A OSPF Router with ID () (Process ID 21) Type-7 AS External Link States (Area 7) LS age: 162
Options: (No TOS-capability, Type 7/5 translation, No DC) LS Type: AS External Link Link State
ID: 172.16.1.140 (External Network Number ) Advertising Router: 100.1.1.1 LS Seq Number:80000012
Checksum: 0x504c Length: 36 Network Mask:255.255.255.255 Metric Type: 1 (Comparable directly to
link state metric) TOS: 0 Metric: 20 Forward Address: 192.168.1.1 --- Vlan101 External Route
Tag: 0 N7K# show ip ospf data extroute172.16.1.150 det vrf B OSPF Router with ID () (Process ID
10 VRF B) Type-5 AS External Link States LS age: 323 Options: 0x20 (No TOS-capability, DC) LS
Type: Type-5 AS-External Link State ID:route172.16.1.150 (Network address) Advertising
Router:100.1.1.2 LS Seq Number: 0x8000008f Checksum: 0x9691 Length: 36 Network Mask: /32 Metric
Type: 1 (Same units as link state path) TOS: 0 Metric: 20 Forward Address: 10.1.1.1 --- Vlan 102
External Route Tag: 0
```